Ohm’s law

antocciphysics | TEACHER KEY | CPA

Learning target: Connect voltage, current and resistance for an ohmic component.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 3; 3 course days later.*
Positive charge: left. Negative charge: right. Each force has magnitude 15 N.

02. Explain a worked example (worked)
Closed loop; 12 V across 4 ohms; 3 A conventional current from positive supply terminal through resistor.

03. Look and explain (model)
Straight line through origin; slope V/I is resistance.

04. Build the solution (practice)
2 A.

05. Use the idea (reason)
Current halves because I=V/R.

06. Choose a method (mixed) *Review: Unit 6, Day 12; 12 course days later.*
Same speed c; the higher-frequency wavelength is 1/4 as large.

07. Exit check (exit)
8 V.

08. STRETCH: Apply it another way (stretch)
E.g.(4 V, 2 ohms),(8 V, 4 ohms),(12 V, 6 ohms). To isolate resistance, hold voltage fixed and vary resistance.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 3; 3 course days later.*
Field becomes 1/9; assumes an isolated point charge or an exterior spherically symmetric source.

02. Explain a worked example (worked)
Closed loop; 12 V across 4 ohms; 3 A conventional current from positive supply terminal through resistor.

03. Represent and interpret (model)
Straight line through origin; slope V/I is resistance.

04. Solve and justify (practice)
30 Ω; .30 A.

05. Analyze the reasoning (reason)
Current halves because I=V/R.

06. Choose a method (mixed) *Review: Unit 6, Day 12; 12 course days later.*
λ≈0.05 m; microwave.

07. Exit check (exit)
20 Ω; V/I is constant for these measurements.

08. STRETCH: Apply it another way (stretch)
R 8 ohms; I 1.25 A. Temperature/material changes can alter resistance.